CT36 | NORMALIZED ELLIPTIC DUAL CHAINS

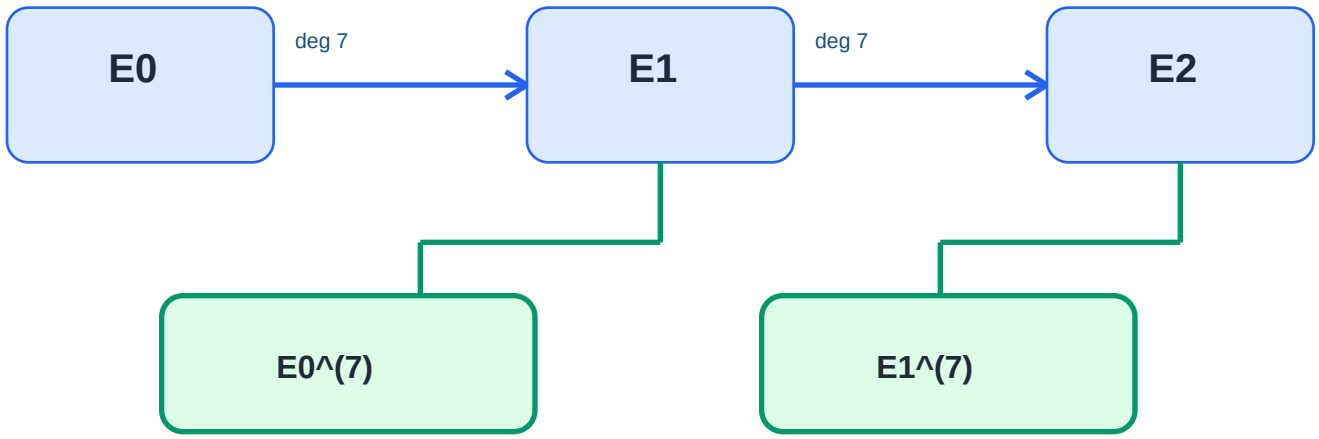
F101 - 2026-11-12 - unsigned developer audit [pcc:unverified]

Explicit reverse degree-7 kernels (low-to-high coefficients)

Chain	Forward curve	Reverse kernel	Dual target
109 / 0	(1,46) -> (26,58)	(58,26,58,1)	(78,72)
109 / 1	(26,58) -> (59,27)	(13,62,81,1)	(8,82)
113 / 0	(44,21) -> (69,50)	(29,88,45,1)	(99,68)
113 / 1	(69,50) -> (23,7)	(11,64,23,1)	(29,8)

Each reverse codomain is E^(7): (a,b) -> (7^4*a,7^6*b).

Its normalized map equals the classical dual followed by s_7.



Symbolic (universal): psi_i o phi_i = s7 o [7] in F101(x).

Two-hop inverse: Psi o Phi = s49 o [49].

4 exact rational polynomial identities; 49,300 exhaustive reverse pairs.

222/222 source points pass the 49-multiplication roundtrip.

Wrong reverse cubics rejected by division-polynomial and identity guards.

Scope: elliptic factors only; Jacobian maps, independent PCC, cost gates open.

Source: cryptanalysis PR CT36; classical dual isogeny (Silverman III.6).